Uncertainty-Aware Risk-Adjusted Benchmarking of the Fama–French Portfolios: Latent Performance, Rank Uncertainty, and the Limits of Forecastability

Muhammad Shoaib

Working paper · This version: 16 July 2026

Abstract

We benchmark the 25 Fama–French size and book-to-market portfolios on a validated balanced panel of 29,825 portfolio-months (1,193 months, July 1926–November 2025). The estimand is the Fama–French three-factor (FF3) alpha; the primary uncertainty object is the full 25×25 heteroskedasticity- and autocorrelation-consistent (HAC) covariance of the alpha vector; and the reported score is a multivariate empirical-Bayes posterior alpha. Rankings are accompanied by common-date block-bootstrap rank intervals and evaluated out of sample. The joint HAC/Wald test rejects the hypothesis that all alphas are zero (p = 6.42e-10), but partial pooling, wide bootstrap rank intervals, and strong long-horizon mobility show that the point ranking is not a precise hierarchy. In strictly forward windows the top-minus-bottom quintile spread averages 243 bps/year, yet an incremental test decomposes this signal: the time-varying rolling ranking adds no predictive power over a leakage-free expanding-window ranking (mean increment -0.042, HAC one-sided p = 0.97) and is dominated by a single fixed full-sample ordering. The forward result is therefore consistent with persistent characteristic mispricing rather than dynamic forecasting skill. Alpha is benchmark-dependent (an 11-rank FF3-to-FF5 move), and only one of 25 stochastic-frontier boundary tests survives false-discovery control.

Keywords: cross-sectional asset pricing; empirical Bayes; HAC inference; rank uncertainty; out-of-sample validation; false-discovery control.

JEL classification: G11, G12, C11, C58.

1. Introduction

A portfolio can earn a high average return because it bears systematic risk, because its sample happened to be favourable, because the benchmark is incomplete, or because it contains genuinely unusual performance. Factor benchmarking removes the part linearly associated with priced factors and studies the intercept, or alpha. For the 25 Fama–French portfolios—the canonical test assets from which the size and value factors are built (Fama and French, 1993)—this intercept is a model-relative quantity: it is unexplained by the chosen factors, not by every conceivable risk or friction.

This paper makes the uncertainty around such a benchmark explicit and then asks a sharper question than the usual out-of-sample test. Our contributions are fourfold. First, we estimate the full cross-portfolio HAC covariance of the 25 alphas (Newey and West, 1987) rather than 25 unrelated standard errors, and use it for joint testing and for multivariate empirical-Bayes shrinkage (Efron and Morris, 1973). Second, we quantify ranking uncertainty with a common-date circular block bootstrap (Politis and Romano, 1992) that refits every stage of the estimator. Third, we separate description from validation with a strictly forward evaluation in which scores, ranks, and factor loadings are frozen before the evaluation window. Fourth—and centrally—we introduce an *incremental* forward test that isolates whether the time-varying ranking adds anything beyond a fixed characteristic ordering.

The headline results are as follows. The joint zero-alpha hypothesis is rejected under the dependence-robust HAC/Wald test, and 5 of 25 individual alphas survive 5% Benjamini–Hochberg false-discovery control (Benjamini and Hochberg, 1995). Yet bootstrap rank intervals are wide and long-horizon quintile mobility is high, so the point ranking is not a reliable league table. The strictly forward top-minus-bottom quintile spread is positive on average (243 bps/year). The incremental test then shows that this predictability is *not* dynamic: the 120-month rolling ranking beats a naive size/value characteristic gradient (increment +0.116) but does not beat a leakage-free expanding-window ranking (-0.042, p = 0.97), and is itself dominated by a single fixed full-sample ordering (-0.100). The predictive content lives in a stable, irregular cross-sectional alpha pattern that FF3 misprices persistently, not in the re-estimation of the ranking through time.

The remainder of the paper is organised as follows. Section 2 describes the data and integrity checks. Section 3 sets out the estimator: the factor regression, the joint HAC covariance, empirical-Bayes shrinkage, joint and multiple testing, the bootstrap, and the incremental forward design. Section 4 reports the empirical results, Section 5 the robustness and validation analysis, Section 6 discusses interpretation and limitations, and Section 7 concludes.

2. Data

2.1 Portfolios and sample

The 25 test portfolios form a 5×5 grid. At each formation date eligible stocks are sorted independently into five size and five book-to-market groups; returns within each of the 25 intersections are value weighted. Labels such as SMALL HiBM and BIG LoBM describe the two sorting dimensions, not individual firms or managed funds. The unit of observation is one portfolio-month; the dependent variable is the portfolio return minus the one-month risk-free rate, in decimal monthly units. The canonical benchmark is FF3 (Mkt–RF, SMB, HML). Annualised basis points multiply monthly decimal alpha by 12 and by 10,000, a linear reporting convention rather than geometric compounding.

Table 1. Sample and canonical specification.

Element	Specification	Element	Specification
Cross-section	25 size x B/M portfolios	Primary HAC lag	12 months
Time span	1926:07-2025:11	Rolling window	120 months, step 12
Months	1,193	Forward horizon	12 months
Observations	29,825	Bootstrap draws	5,000 (seed 2026)
Benchmark	FF3: Mkt-RF, SMB, HML	Block length	12 months, circular

2.2 Integrity and provenance

Data integrity is treated as part of the specification. The analytical grain is exactly one portfolio-month, giving 29,825 unique keys with zero duplicates, a balanced panel, and no missing calendar months; the loader fails closed on any violation. Each run records the source byte count and SHA-256 digest of both inputs.

A current (May 2026) official snapshot from the Kenneth R. French Data Library is retained separately and never overwrites the fixed local inputs. The local sample ends in November 2025, but its exact acquisition vintage is unverified, and the two sources differ over their overlap—by up to 1.4167 percentage points for portfolio returns and 0.340 percentage points for FF3/RF fields. Official historical revisions are a plausible but unproven cause. Both comparisons are therefore classified NOT_COMPARABLE (visible, non-blocking) rather than PASS. This is a genuine limitation: the headline numbers rest on inputs that cannot currently be reconstructed byte-for-byte from the public source. All within-pipeline integrity and independent-replication checks pass (Section 5.4).

3. Methodology

3.1 Factor model and the meaning of alpha

For portfolio i in month t, with excess return $y_{it} = r_{it} - r_{ft}$, the benchmark regression is

$$y_{it} = \alpha_i + \beta_i' f_t + \varepsilon_{it},$$

where f_t stacks the FF3 factor returns and α_i is the mean excess return left after the fitted linear factor component is removed. Ordinary least squares gives $\hat{\theta}_i = (X'X)^{-1}X'y_i$. Selecting the intercept, the estimation error is a weighted sum of residual shocks, $\hat{\alpha}_i - \alpha_i = \sum_t a_t \varepsilon_{it}$, where the scalar weight a_t depends only on the common factor design. Because every portfolio shares the same dates and factors, these weights combine with the same-month residual vector to yield a *joint* covariance. A positive alpha denotes in-sample benchmark-relative outperformance, not managerial skill or future profit; a negative alpha can reflect omitted risk rather than inefficiency.

3.2 Full joint HAC covariance

Let e_t be the 25-vector of same-month residuals and $g_t = a_t e_t$ the alpha score vector. With L = 12 lags, Bartlett weights $w_l = 1 - l/(L+1)$, and lag-l cross-product $\Gamma_l = \sum_{t>l} g_t g_{t-l}'$, the covariance is

$$V_{HAC} = [T/(T-P)]\,\{\Gamma_0 + \sum_{l=1}^{L} w_l(\Gamma_l + \Gamma_l')\},$$

with $P = K + 1$ regression coefficients. This 25×25 matrix keeps individual alpha variances on its diagonal and cross-portfolio co-movement off-diagonal. The implementation symmetrises the result, checks eigenvalues, and permits only scale-relative flooring of negligible numerical negatives. The canonical matrix is positive definite without flooring, with condition number 139.4. HAC inference is robust to the modelled dependence; it does not make residuals Gaussian or repair an incomplete benchmark.

3.3 Multivariate empirical Bayes

Ranking 25 noisy estimates invites a winner's curse. Partial pooling treats the latent alphas as one cross-section,

$$\hat{\alpha} \mid \alpha \sim N(\alpha, V_{HAC}), \quad \alpha \sim N(\mu 1, \tau^2 I),$$

with posterior mean $\mu 1 + \tau^2(V_{HAC} + \tau^2 I)^{-1}(\hat{\alpha} - \mu 1)$ and conditional covariance $\tau^2 I - \tau^4(V_{HAC} + \tau^2 I)^{-1}$. The hyperparameters (μ, τ) are estimated by profile marginal maximum likelihood, so the prior is learned from the same cross-section. The fitted prior mean is -26 bps/year with dispersion 94 bps/year, implying meaningful shrinkage toward a slightly negative centre; shrinkage moves at least one portfolio by 10 ranks. Because V_{HAC} is non-diagonal, shrinkage is multivariate. The reported intervals include first-order uncertainty in the profiled mean but do not fully integrate uncertainty in τ, so they are model-based summaries rather than exact finite-sample guarantees.

3.4 Joint and multiple testing

The primary joint test is the dependence-robust Wald statistic $W = \hat{\alpha}' V_{HAC}^{-1} \hat{\alpha} \sim \chi^2(25)$ under the null that all alphas are zero. The Gibbons–Ross–Shanken F-test (Gibbons, Ross, and Shanken, 1989) is reported as a conventional secondary benchmark; its exact F reference assumes IID Gaussian errors, which the diagnostics reject (Section 5.3). Individual raw-alpha p-values and the 25 SFA boundary p-values form two separate Benjamini–Hochberg families, since they test different nulls.

3.5 Bootstrap rank uncertainty

A rank is discontinuous, so individual standard errors do not describe how often a portfolio stays in the top five. We draw 5,000 circular 12-month blocks of common date indices, apply the same dates to all 25 portfolios (preserving cross-sectional shocks), and refit the regressions, the full HAC covariance, and the empirical-Bayes hyperparameters within every replicate before re-ranking. Rank and score intervals are empirical 2.5–97.5% quantiles of this resampling distribution. Increasing the draw count from 1,000 to 5,000 shifts rank interval endpoints by at most 1 rank but alpha-tail endpoints by up to 27 annualised bps, motivating the larger final count.

3.6 Forward validation and the incremental test

For each 120-month training window we freeze all scores, ranks, prior parameters, and betas at the cutoff, then compute each portfolio's next 12-month benchmark-adjusted return with the frozen loadings, $A_i^+ = \mathrm{mean}_t(y_{it} - \hat{\beta}_i' f_t)$. Within each evaluation year we compute the Spearman correlation between training scores and future alpha and the average top-minus-bottom quintile spread. Annual statistics are aggregated with a Fisher transformation and four-lag HAC inference.

The forward test alone cannot separate genuine dynamic skill from persistent characteristic mispricing: because FF3 misprices fixed characteristic cells in a stable way, a ranking anchored to past alpha will correlate with future alpha mechanically. We therefore add an *incremental* test that scores the rolling ranking against four fixed or leakage-free benchmark orderings on the *same* forward observations, and tests whether the rolling ordering adds power. The benchmarks are (i) a size-ascending / B/M-descending characteristic gradient; (ii) a diagonal value-minus-size gradient; (iii) a leakage-free *expanding-window* ranking re-estimated on all data up to each cutoff; and (iv) a full-sample *oracle* ordering (which uses look-ahead and is thus an upper bound on what a static ranking can achieve). Each per-window increment (rolling minus benchmark) is aggregated with an automatic-bandwidth Newey–West HAC t-test and a block bootstrap. Benchmark (iii) is the fair, tradable comparator on which the interpretation rests.

4. Results

4.1 Cross-sectional posterior performance

Table 2 reports the full ranking. SMALL HiBM has the highest posterior alpha (132 bps/year) and SMALL LoBM the lowest (-185 bps/year), reproducing the well-known small-value premium and small-growth shortfall relative to FF3. 2 posterior intervals are wholly above zero (SMALL HiBM, BIG LoBM); most intervals cross zero. Five raw-alpha tests survive BH control, but four of these correspond to significantly negative alpha, so raw-test significance and posterior rank are distinct summaries.

Table 2. Posterior FF3 performance ranking (annualised basis points).

Rank	Portfolio	Raw alpha	Post. alpha	Post. 95%	BH q	Boot. rank 95%
1	SMALL HiBM	169	132	[23, 240]	0.063	[1.0, 6.0]
2	BIG LoBM	118	79	[9, 148]	0.033	[1.0, 10.0]
3	ME4 BM1	50	78	[-24, 180]	0.732	[1.0, 15.0]
4	ME1 BM4	84	61	[-47, 169]	0.456	[1.0, 12.0]
5	ME3 BM2	103	55	[-43, 153]	0.227	[1.0, 12.0]
6	ME2 BM5	37	46	[-49, 141]	0.766	[2.0, 15.0]
7	ME4 BM4	23	21	[-83, 125]	0.861	[2.0, 18.0]
8	ME2 BM2	21	16	[-92, 125]	0.868	[2.0, 18.0]
9	ME3 BM4	44	14	[-80, 107]	0.732	[3.0, 18.0]
10	ME5 BM3	-38	13	[-95, 121]	0.825	[3.0, 19.0]
11	ME2 BM3	26	-7	[-105, 92]	0.825	[3.0, 19.0]
12	ME2 BM4	25	-17	[-104, 70]	0.825	[4.0, 19.0]
13	ME5 BM2	-5	-29	[-112, 55]	0.912	[5.0, 20.0]
14	ME4 BM3	-15	-29	[-130, 73]	0.874	[6.0, 21.0]
15	ME3 BM3	52	-35	[-128, 59]	0.732	[4.0, 20.0]
16	ME3 BM5	-76	-43	[-155, 68]	0.562	[6.0, 22.0]
17	ME4 BM2	-14	-77	[-184, 31]	0.874	[9.0, 23.0]
18	BIG HiBM	-198	-82	[-229, 65]	0.336	[10.0, 24.0]
19	ME3 BM1	-142	-82	[-178, 15]	0.055	[11.0, 23.0]
20	ME1 BM3	-144	-83	[-192, 25]	0.106	[9.0, 23.0]
21	ME4 BM5	-211	-92	[-218, 35]	0.063	[13.0, 24.0]
22	ME1 BM2	-476	-115	[-271, 41]	0.005	[15.0, 25.0]
23	ME2 BM1	-261	-115	[-229, -1]	0.003	[15.0, 25.0]
24	ME5 BM4	-289	-164	[-268, -60]	0.002	[19.0, 25.0]
25	SMALL LoBM	-810	-185	[-351, -18]	<0.001	[21.0, 25.0]

Note: Raw and posterior alpha in annualised basis points; posterior 95% is the empirical-Bayes interval; BH q is the false-discovery-adjusted raw-alpha p-value; the bootstrap rank interval is the central 95% range over 5,000 common-date circular-block draws.

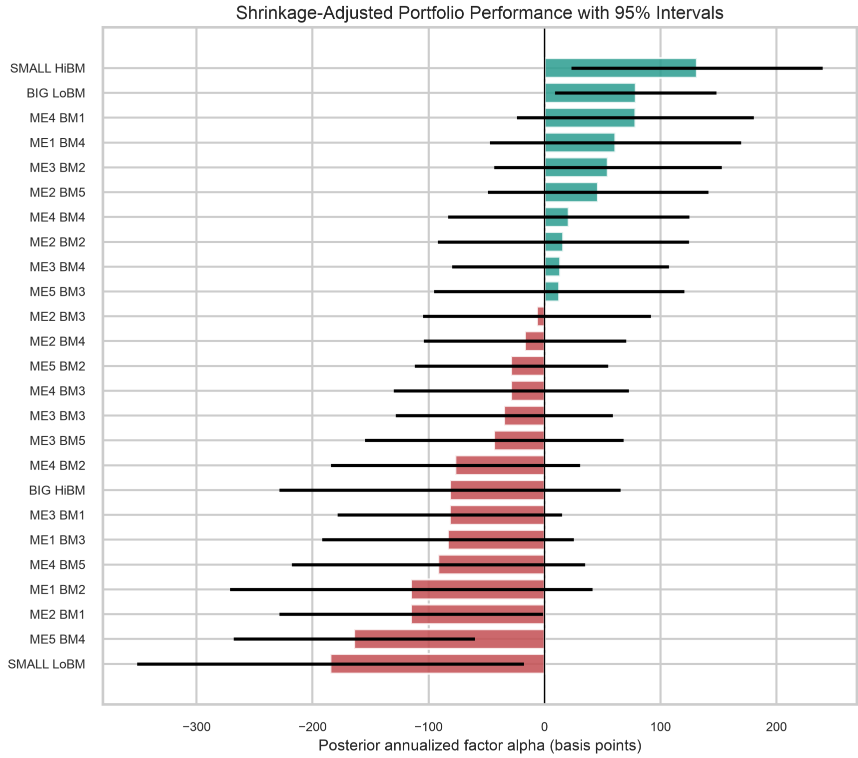

Figure 1. Posterior FF3 alpha (annualised bps) with approximate 95% posterior intervals. An interval crossing zero indicates the latent alpha is not cleanly separated from zero; interval overlap warns that adjacent ranks are not distinguished.

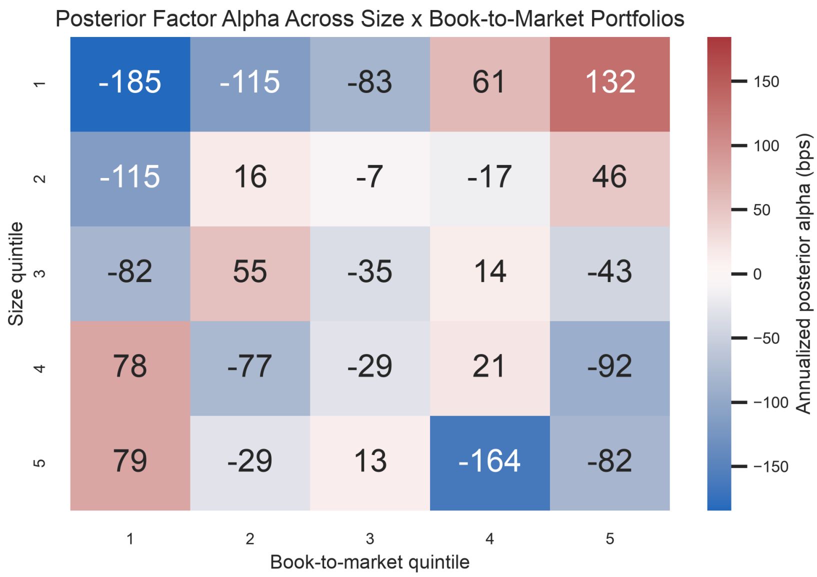

Figure 2. Posterior annualised alpha on the 5×5 size/book-to-market grid. The grid restores the economic sorting structure that a one-dimensional list hides and should be read jointly with intervals and benchmark sensitivity.

4.2 Joint tests

Table 3. Joint tests of the hypothesis that all 25 alphas are zero.

Test	Statistic	p-value	Role
HAC/Wald	93.95	6.42e-10	Primary; dependence-robust
GRS	3.28	1.16e-07	Secondary; IID-Gaussian reference

Both tests reject. Rejection means the 25-dimensional alpha vector is inconsistent with all-zero under the stated model; it does not establish that alphas are positive, that the ordering is stable, or that any portfolio reflects persistent skill.

4.3 Rank uncertainty

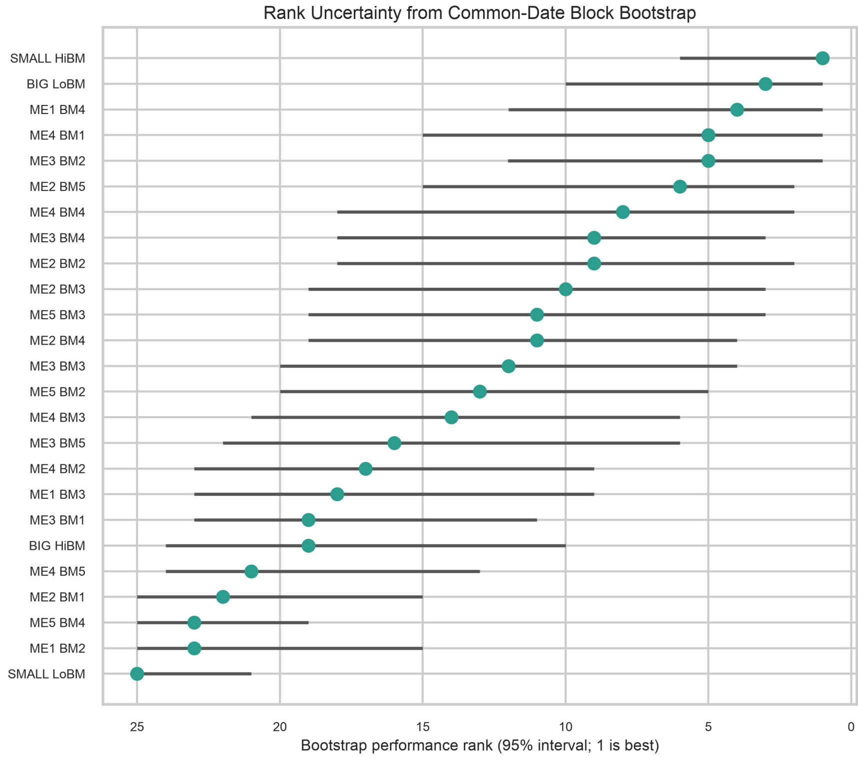

Figure 3. Median bootstrap rank and central 95% rank interval from 5,000 common-date circular-block draws. Broad, overlapping intervals are direct evidence against reading the ordered list as a precise hierarchy.

SMALL HiBM is rank 1 at the point estimate and appears in the top quintile in 97% of bootstrap histories, yet its central rank interval still reaches rank 6. Rank intervals overlap broadly across portfolios.

4.4 Persistence and mobility

Table 4. Rank and score persistence by calendar horizon.

Horizon	Window overlap	Mean rank rho	Mean score r	Mean |rank chg|	Structural
12m	90%	0.863	0.892	2.67	No
60m	50%	0.483	0.531	5.62	No
120m	0%	0.156	0.197	7.44	Yes

Rank persistence falls from about 0.86 at 12 months (90% window overlap) to roughly 0.16 at 120 months (zero overlap). The non-overlapping horizon is the credible structural measure and is weak: the mean probability of staying in the same quintile after 120 months is 25% (Table 5), only modestly above the 20% implied by uniform mobility.

Table 5. Quintile transitions over a non-overlapping 120-month horizon.

Starting quintile	Stay	Move up	Move down
1	28%	72%	0%
2	22%	54%	24%
3	22%	43%	34%
4	23%	20%	57%
5	28%	0%	72%

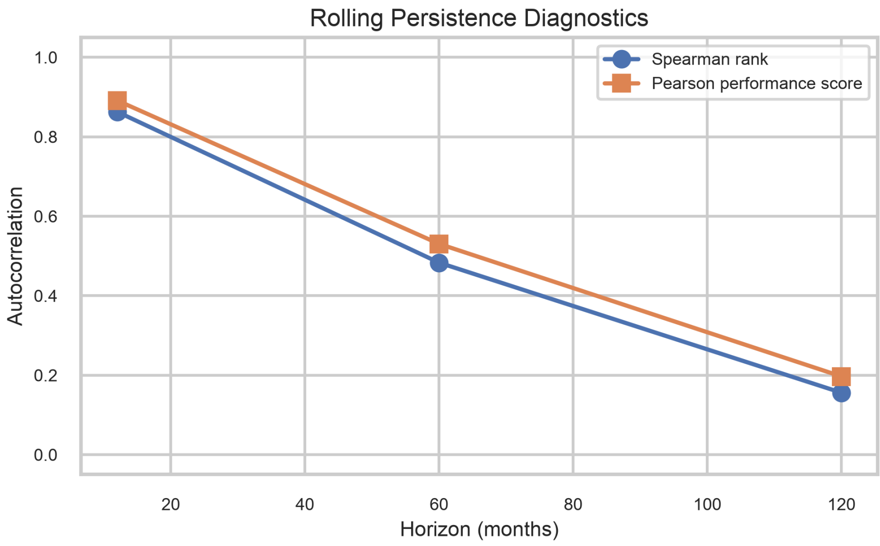

Figure 4. Mean rank and score persistence by calendar horizon, with window overlap shown explicitly. The 120-month point compares disjoint decades.

4.5 Strictly forward validation

Across 89 non-overlapping 12-month evaluation periods, the Fisher-averaged rank correlation between training scores and future alpha is 0.102 (HAC 95% CI [0.055, 0.149]), and the average top-minus-bottom quintile spread is 243 bps/year (median 209, IQR 498), positive in 73% of windows (Table 6). Both means are statistically distinguishable from zero, but dispersion across years is large—the IQR exceeds the mean—so the average is not a uniformly reliable timing rule.

Table 6. Strictly forward validation (four-lag HAC inference).

Metric	Estimate	HAC 95% interval	HAC p
Rank vs future alpha	0.102	[0.055, 0.149]	2.32e-05
Top-bottom quintile (bps/y)	243	[153, 333]	1.30e-07

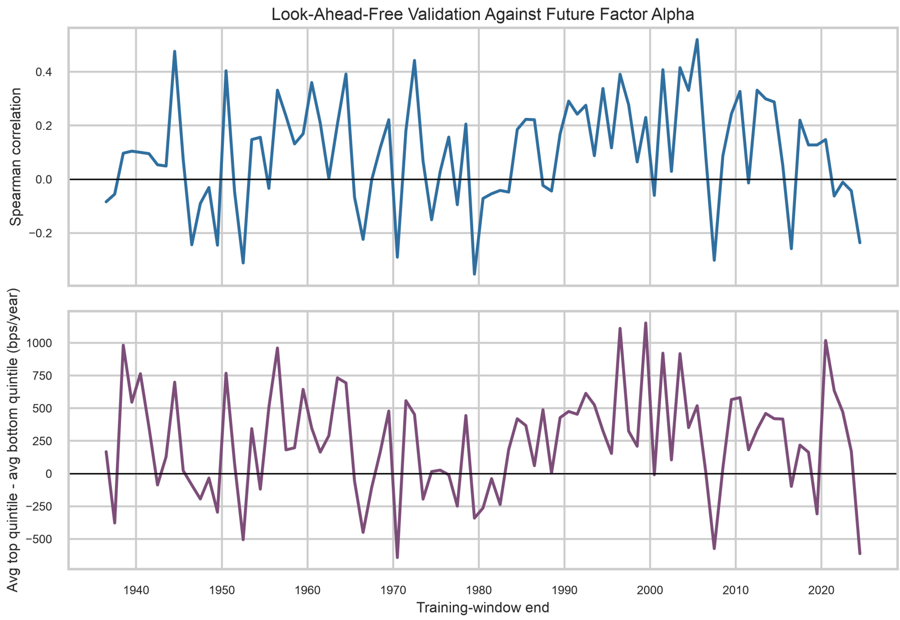

Figure 5. Window-by-window future rank correlation and average top-minus-bottom quintile future alpha. Frequent negative windows and large spread variation show the average is not uniformly reliable.

4.6 The incremental test: is the ranking dynamically informative?

Table 7 decomposes the forward signal. The rolling ranking's own level correlation with future alpha is +0.099. It exceeds the two static characteristic gradients—which have essentially zero forward power, because FF3 alpha is the residual after size and value exposure are removed—so the rolling signal is not merely a restated size/value tilt. But it does not beat the leakage-free expanding-window ranking (level +0.140; increment -0.042, HAC one-sided p = 0.97), and it is dominated by the full-sample oracle ordering (level +0.199; increment -0.100, significantly negative). A single fixed but irregular cross-sectional ordering predicts future alpha as well as or better than the ranking that is re-estimated every year.

Table 7. Incremental predictive power of the rolling ranking versus fixed and leakage-free benchmarks.

Benchmark ordering	Bench. corr	Rank-corr incr.	HAC p	Bootstrap 95% CI	Spread incr.
Characteristic gradient (size asc, B/M desc)	-0.018	+0.116	<0.0001	[+0.062, +0.169]	+343
Characteristic gradient (diagonal)	+0.006	+0.092	0.0004	[+0.038, +0.145]	+221
Expanding window (leakage-free)	+0.140	-0.042	0.9691	[-0.085, +0.001]	-44
Full-sample alpha (oracle, look-ahead)	+0.199	-0.100	1.0000	[-0.148, -0.053]	-165

Note: The rolling ranking's own level correlation with future alpha is +0.099. Δ rank corr is the mean per-window Spearman increment (rolling minus benchmark); HAC p is the one-sided test that the increment exceeds zero; Δ spread is the mean top-minus-bottom spread increment in annualised bps. Benchmarks are ordered from naive (characteristic gradients) to strong (expanding window; full-sample oracle). The oracle uses look-ahead information and is an upper bound, not a tradable strategy.

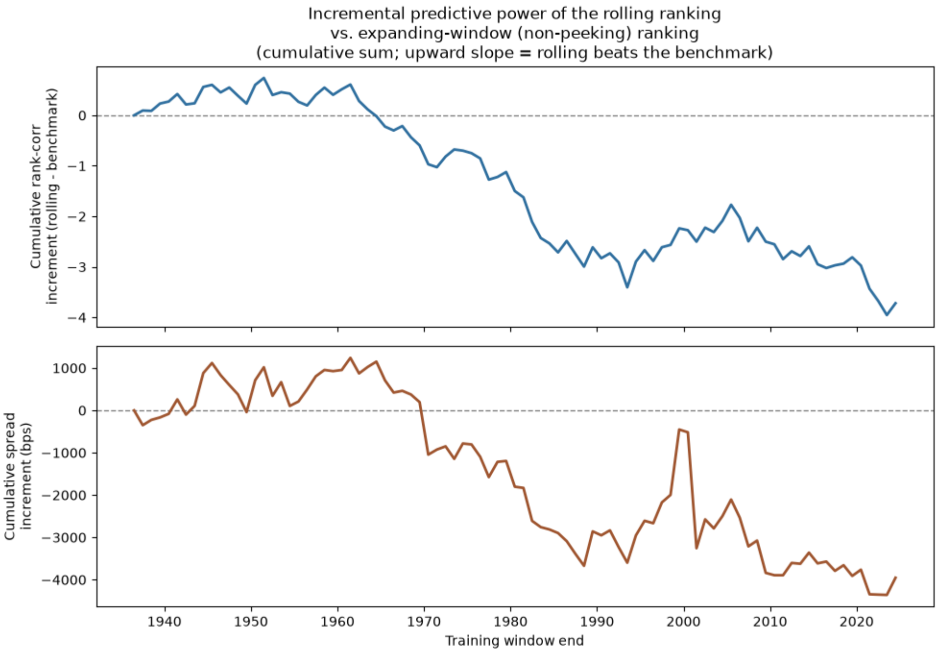

Figure 6. Cumulative rolling-minus-expanding-window increment in rank correlation (top) and quintile spread (bottom). A flat or downward path shows the time-varying ranking does not out-predict a ranking that simply accumulates all past data.

The interpretation is unambiguous. The out-of-sample predictability documented in Section 4.5 is real but reflects a stable, irregular pattern of characteristic mispricing that a fixed ranking captures at least as well; discarding data older than the rolling window and chasing recent alpha adds estimation noise, not forecasting skill. Claims of *dynamic* forecasting ability are not supported by the data.

5. Robustness and validation

5.1 Benchmark sensitivity: FF3 versus FF5

On the prespecified common sample (1963:07–2025:11), FF3 and FF5 posterior ranks have Spearman correlation 0.875. ME5 BM2 moves from rank 9 under FF3 to rank 20 under FF5, an 11-position change. FDR-supported raw alphas move from 5 under FF3 to 3 under FF5, and the HAC/Wald test rejects joint zero alpha under both FF3 (p = 1.63e-08) and FF5 (p = 6.67e-07). The 11-rank maximum movement demonstrates that alpha is benchmark-dependent: part of what looks like FF3 performance is exposure to dimensions absent from FF3.

5.2 Window-length sensitivity

Table 8. Rolling-window-length sensitivity of the ranking.

Windows compared	Rank corr	Score corr	Top Jaccard	Bottom Jaccard
60m vs 120m	0.717	0.746	0.393	0.367
60m vs 180m	0.546	0.580	0.325	0.294
120m vs 180m	0.782	0.802	0.515	0.448

Rank correlations across 60-, 120-, and 180-month windows are positive but far from one, with top-quintile membership overlap (Jaccard) that can fall below 0.40. Window length is not a neutral tuning knob; disagreement reflects genuine temporal instability.

5.3 Residual diagnostics and model risk

Residual normality is rejected for 25 of 25 portfolios (Jarque–Bera), serial independence for 16 of 25 (Ljung–Box, lag 12), and conditional homoskedasticity for 25 of 25 (ARCH-LM, lag 12). An IID-Gaussian error model is not credible for these returns, which is why the paper relies on HAC inference and block resampling and treats GRS as secondary. These estimators protect parts of the inference against dependence; they do not repair omitted nonlinearities, time-varying betas, or benchmark incompleteness.

The historical 1963–1991 anchor sharpens the point: HAC/Wald rejects ($p = 0.0003$) while GRS does not ($p = 0.0821$), consistent with the diagnostic evidence against IID-normal residuals.

5.4 Independent replication and reference validation

All coefficients and HAC standard errors reproduce independently with statsmodels to a tolerance of 1e-12 for 25 of 25 portfolios, and the qualitative 5×5 size/value patterns, excess-return identity, block selection, and weighting checks all pass. Independent numerical replication is distinct from source reconciliation: the two current-snapshot comparisons remain NOT_COMPARABLE (Section 2.2) because the local acquisition vintage is unverified, and no external comparison whose values differ is reported as a pass.

5.5 Stochastic frontier analysis as a narrow diagnostic

A half-normal stochastic-frontier model ($\varepsilon = v - u$, $u \geq 0$) is applied only as a residual-asymmetry diagnostic. Because a free intercept and a one-sided shortfall are not separately identified, SFA is not used as a substitute for alpha. The boundary null $\sigma_u = 0$ is tested with the appropriate 50:50 χ^2 mixture and BH-corrected across the 25 portfolios (Jondrow et al., 1982). Only 1 of 25 boundary tests survives 5% FDR (ME5 BM4), so no general efficiency ranking is warranted.

6. Discussion

Table 9 summarises what the evidence does and does not support. The central message is a distinction the raw forward test obscures: a *stable* cross-sectional ordering of the 25 cells carries out-of-sample information, but the *time variation* of the rolling ranking does not. The most defensible use of the framework is comparative and uncertainty-aware—estimate benchmark-relative performance, shrink noisy extremes, and report rank uncertainty. The least defensible use is to select a single portfolio from the point rank and treat its historical posterior alpha, or the rolling ranking's apparent timing ability, as a guaranteed future excess return.

Table 9. What the evidence supports.

Statement	Assessment	Basis
Some FF3 alphas differ from zero	Supported	FDR evidence and joint HAC/Wald rejection
The point rank is a precise hierarchy	Not supported	Wide bootstrap rank intervals; strong mobility
A stable ordering has out-of-sample information	Supported	Positive forward spread; expanding-window and oracle levels
The time-varying (rolling) ranking forecasts	Not supported	No increment over expanding window; dominated by oracle
The forward relation is stable through time	Not supported	Large IQR; negative windows; decade heterogeneity
Alpha is independent of benchmark choice	Not supported	11-rank FF3-to-FF5 movement
SFA yields a full efficiency ranking	Not supported	Only 1 of 25 boundary tests survives FDR
The implementation reproduces independently	Supported	OLS/HAC match statsmodels to 1e-12
Local data equal the latest official snapshot	Not supported	Two NOT_COMPARABLE comparisons; vintage unverified

6.1 Limitations

- Returns exclude transaction costs, taxes, shorting frictions, capacity, and market impact; the top-minus-bottom construction shorts small-growth, which is costly and capacity-constrained in practice.
- The source portfolios are value-weighted research constructions reconstituted by a published methodology, not live investable funds; full-sample alphas embed later data revisions.
- The factor model is linear with constant betas within each window; time-varying exposures, nonlinear payoffs, and omitted priced factors can remain in residuals.
- Posterior intervals do not fully integrate hyperparameter uncertainty, and the full HAC covariance is treated as known in the shrinkage step, so reported intervals are mildly optimistic.
- The oracle benchmark in Table 7 uses look-ahead information and is an upper bound; the expanding-window benchmark is the leakage-free comparator on which the negative incremental verdict rests.
- The local input vintage cannot currently be reconciled with the public source (Section 2.2).

7. Conclusion

This paper provides an uncertainty-aware benchmark for the 25 Fama–French portfolios built on a validated panel, a full joint HAC covariance, multivariate empirical-Bayes shrinkage, multiplicity control, overlap-aware persistence analysis, and strictly forward validation. Cross-sectional FF3 alpha variation is present and jointly significant, but its apparent precision is substantially reduced by partial pooling, bootstrap rank uncertainty, temporal mobility, and benchmark sensitivity.

The paper's main contribution is the incremental forward test. It shows that the positive out-of-sample quintile spread is driven by persistent characteristic mispricing captured by a *stable* ordering, not by dynamic skill in the re-estimated ranking: the rolling ranking adds no predictive power over a leakage-free expanding-window ranking and is dominated by a fixed full-sample ordering. Future work should test whether any residual signal survives realistic turnover and trading costs, whether additional prespecified factors absorb the FF3 residual alphas without specification search, and whether a decision-theoretic allocation over rank probabilities improves on hard top-quintile selection. Any real-time claim should also await reconciliation of the data vintage with the official source.

References

Benjamini, Y., and Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.

Efron, B., and Morris, C. (1973). Stein's estimation rule and its competitors—an empirical Bayes approach. *Journal of the American Statistical Association*, 68(341), 117–130.

Fama, E. F., and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.

Fama, E. F., and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.

Gibbons, M. R., Ross, S. A., and Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.

Jondrow, J., Lovell, C. A. K., Materov, I. S., and Schmidt, P. (1982). On the estimation of technical inefficiency in the stochastic frontier production function model. *Journal of Econometrics*, 19(2–3), 233–238.

Newey, W. K., and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.

Politis, D. N., and Romano, J. P. (1992). A circular block-resampling procedure for stationary data. In *Exploring the Limits of Bootstrap*, Wiley, 263–270.

Kenneth R. French Data Library (2026). 25 portfolios formed on size and book-to-market; Fama–French research factors. Official URLs, access date, hashes, units, and coverage are recorded in results/tables/official_reference_manifest.csv.

Appendix A. Reproducibility

The reviewed environment is CPython 3.13; the deterministic bootstrap uses seed 2026. The recorded end-to-end runtime is 383.4 seconds. Source hashes, settings, and runtime are persisted in results/run_manifest.json so another analyst can identify exactly what was run. The full pipeline is:

python -m pip install -r requirements-dev-lock.txt
pytest -q · ruff check .
MPLBACKEND=Agg python -m analysis.run_all
python -m analysis.export_tables
python reports/generate_report.py

The incremental forward test (Section 4.6) is produced within analysis.run_all and can be regenerated independently with *python -m analysis.incremental_validation --scheme all*, which writes incremental_validation_summary_*.csv, incremental_validation_windows_*.csv, benchmark_ranking_*.csv, and the corresponding figures.

Appendix B. Notation

Symbol	Definition	Symbol	Definition
y_it	Excess return r_it - r_ft	V_HAC	Full 25x25 alpha covariance
f_t	Factor return vector	mu, tau	EB prior mean and dispersion
alpha_i	FF3 regression intercept	L	HAC lag length (L = 12)
beta_i	Factor loadings	W	HAC/Wald joint statistic
eps_it	Regression residual	A_i^+	Frozen-beta forward alpha

All numerical claims in this paper are generated from the canonical result tables listed in Appendix A.